

IPsec Security Assessment Report

AES128_SHA256_DH14_NOPFS_ICMP_rep01 | TECHNICAL | 2026-09-27T16:55:09.640Z

Assessment scores

Security 54/100 | Observed risk 10/100 | Evidence coverage 60% (PARTIAL)

Evidence-adjusted score: 54%



Evidence coverage: 60%



Technical interpretation

The capture contains 40 packets totaling 6,800 bytes with a mean packet length of 170 bytes and zero standard deviation, indicating uniform packet sizes. The mean inter-arrival time is 2.6 ms with a burst ratio of 0.9 and flow symmetry of 0.99, suggesting a steady, bidirectional flow. Calculated entropy is 7.752, consistent with encrypted traffic. The IKEv2 negotiation specifies AES-128-CBC for encryption and AUTH-HMAC-SHA2-256-128 for integrity, with DH Group 14 (2048-bit MODP). PFS, key lifetime, and replay protection parameters are not present in the capture, leaving these aspects unassessed. The operational mode is not determined from the capture. No plaintext or application-layer data is visible due to encryption.

Security and risk

Evidence-adjusted security score: 54/100 (Weak).

Observed configuration risk score: 10/100. This is a rule-based penalty, not the probability of an attack.

Evidence coverage: 60% (PARTIAL). Unknown controls receive no security credit.

Threat matrix and remediation

Severity	Control / threat	Observed	Penalty	Action
Low	Symmetric Encryption Cipher: Cipher Block Chaining Mode Padding Attacks	AES-CBC	10	Prefer Authenticated Encryption with Associated Data (AEAD) modes like AES-GCM or ChaCha20-Poly1305.

Evidence gaps

Perfect Forward Secrecy (PFS): Not observed in capture. Capture CREATE_CHILD_SA exchanges or provide an explicit Child SA configuration.

Anti-Replay Window Protection: Not observed in capture. Provide Child SA configuration or negotiated ESN evidence.

Cryptographic parameters

Parameter	Observed value
IKE version	IKEv2
Operating mode	Not determined from capture
Encryption	AES-CBC
Integrity	AUTH-HMAC-SHA2-256-128
DH group	DH Group 14 (MODP 2048-bit)
PFS	Not observed
Replay protection	Not observed
Key lifetime	Not observed

Traffic analysis and metadata inference

Inferred workload: VoIP / Audio Call.

Traffic pattern match: 60%. This is a relative heuristic score, not calibrated model confidence.

ESP packets: 40; mean length: 170.0 bytes; mean inter-arrival: 2.6 ms; entropy: 7.75 bits/byte.

Wire metadata: 6 IKE and 40 ESP frames; NAT traversal Not detected.

Application identity and encrypted contents cannot be confirmed from packet metadata.

AI confidence

AI confidence score: 93%.

Mean predicted-class probability across 4 available cryptographic inference models; this is not measured accuracy.

Cryptographic model predictions

Target	Prediction	Predicted-class probability
encryption	AES128	73%
hash	SHA256	100%
dh group	DH14	100%
pfs group	NOPFS	100%

Model-inferred security notes

These are predictions, not packet-observed configuration findings. Confirm them with gateway telemetry or negotiation evidence.

Severity	Finding	Basis	Confidence	Recommended check
low	Encryption: AES-128 (ML prediction)	ml inferred	73%	Consider upgrading to AES-256 for highest-assurance deployments.
info	Integrity: SHA-256 (ML prediction)	ml inferred	100%	Consider SHA-384 for highest-assurance environments.
medium	Key Exchange: DH Group 14 / MODP-2048 (ML prediction)	ml inferred	100%	Upgrade to DH Group 15 (MODP-3072) or DH Group 19/20 (ECDH).
high	Perfect Forward Secrecy: Disabled (ML prediction)	ml inferred	100%	Enable PFS using CREATE_CHILD_SA with a new DH exchange (DH Group 14 minimum, Group 15+ recommended).
info	IKE Handshake Traffic Observed	observed	100%	IKE control-plane visibility is expected. Ensure IKE_AUTH payloads use encryption (payload type 46).
info	ESP Data Tunnel Active	observed	100%	Verify ESP replay window size is configured appropriately.

Wire-visible metadata

Total frames: 46; IKE: 6; ESP: 40; AH: 0.

NAT traversal: Not detected; capture duration: 30054.0 ms.

IKE exchanges: 34, 35, 37.